

**This assignment will not be graded and is only for practice.**

1) Consider a map  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined as  $T(v) = Av$ , where  $v = \begin{bmatrix} x \\ y \end{bmatrix}$ ,  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  **1 point**  
and  $\det(A) = 0$ . Which of the following options are correct?

- ☐  $T$  is a linear transformation.
- ☐  $T$  is both one-one and onto.
- ☐  $T$  is neither one-one nor onto.
- ☐  $T$  is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T$  is a linear transformation.

$T$  is both one-one and onto.

### Recall the definition of a linear transformation

- A function  $T$  from one vector space  $V$  to another vector space  $W$  ( $T : V \rightarrow W$ ) is said to be a linear transformation if for any two vectors  $v_1$  and  $v_2$  in the vector space  $V$  and for any  $c \in \mathbb{R}$  (scalar) the following conditions hold:
  - $T(v_1 + v_2) = T(v_1) + T(v_2)$
  - $T(cv_1) = cT(v_1)$

### Step 1:

2) What is  $T(v_1 + v_2)$ ?

**1 point**

- ☐  $A(v_1 + v_2)$ .
- ☐  $Av_1v_2$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A(v_1 + v_2)$ .

**Recall:**  $A(v_1 + v_2) = Av_1 + Av_2$

**Step 2:**

3) Is  $T(v_1 + v_2) = T(v_1) + T(v_2)$ ?

**1 point**

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

**Check:**  $T(cv) = cT(v)$ .

Hence  $T$  is linear transformation. So Option 1 is true.

**Another method:** Write the definition of  $T$  explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$

**Step 1:** What is  $T((x_1, y_1) + (x_2, y_2))$ ?

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) \\ &= T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

**Step 2:**

4) Is  $T(c(x, y)) = cT(x, y)$ ?

**1 point**

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

Yes.

Hence  $T$  is linear transformation. So Option 1 is true.

**Recall the definition of a one to one and an onto linear transformation  $T : V \rightarrow W$**

- If  $T(v_1) = T(v_2)$  implies  $v_1 = v_2$ , then  $T$  is one to one.
- If for each  $w \in W$  there exists  $v \in V$  such that  $T(v) = w$ , then  $T$  is onto.
- $T(v) = 0$  implies  $v = 0$ , if and only if  $T$  is one to one.
  - Suppose  $T$  is one to one. Now let us assume that  $T(v) = 0$ . We also have  $T(0) = 0$ . Hence,  $T(v) = T(0)$ . As  $T$  is one to one, by the definition of a one to one transformation, we have  $v = 0$ .
  - Conversely assume that  $T(v) = 0$  implies  $v = 0$ . We will show  $T$  is one to one. Let  $T(v_1) = T(v_2)$ . So we have,  $T(v_1 - v_2) = T(v_1) - T(v_2) = 0$ . Hence  $v_1 - v_2 = 0$ , which implies  $v_1 = v_2$ . So  $T$  is one to one.

Now in the given problem  $T(v) = 0$ , implies  $Av = 0$ . This is the matrix representation of the system of linear equations:

$$ax + by = 0$$

$$cx + dy = 0$$

## Step 3:

5) If the system of linear equations  $Av = 0$  has a unique solution, then what is the solution?

**1 point**

- ☐  $v = 0$
- ☐  $v$  can be any vector in  $\mathbb{R}^2$ .

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$v = 0$

**Step 4:**

6) What is the condition on  $A$ , for which the system of linear equations  $Av = 0$  has a **1 point** unique solution? [Recall from Weeks 1 and 2]

☐  $\det(A) = 0$ .

☐  $\det(A) = 0$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A) = 0$ .

Concluding, we can say that, if  $\det(A)$  is non-zero, then  $T(v) = 0$  implies  $v = 0$ . Hence  $T$  is one to one.

Let  $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in \mathbb{R}^2$ . We are going to find whether there exists a vector  $v = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ , such that

$T(v) = Av = w$ .

$Av = w$  is the matrix representation of the system of linear equation:

$$ax + by = w_1$$

$$cx + dy = w_2$$

**Step 5:**

7) If  $\det(A) = 0$ , then what can we say about the solution of the above system of linear **1 point** equations?

☐ No solution.

☐ Unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Unique solution.

Concluding from this, we can say that, if  $\det(A) = 0$ , then  $T$  is onto.

**Hence, Options 1 and 2 are the correct options.**

Your score is: 0/7

